

An Integrated 150 Gb/s NRZ Heterodyne Detection RX Supporting 20km SMF with Simple FFE for VHSP PON

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Abstract An integrated NRZ modulation heterodyne detection RX is demonstrated for VHSP PON. Analog signal processing for chromatic dispersion suppression and demodulation is implemented. 150 Gb/s NRZ is detected with -22(B2B)/-16dBm(20km SMF) OMA at 1320/1295nm, using 0.5 dBm LO and 7/11-tap FFE. ©2026 The Author(s)

Introduction

With rapid development of AI and 6G infrastructure, the demand for very high speed (VHSP) PON emerges. The exploration of 100G PON under the framework of very high speed PON (VHSP) in ITU-T is in progress. To meet the specifications of high data rate, long fiber reach, and optical budget, different technical schemes have been studied. Conventional intensity modulation direct detection (IMDD) schemes using avalanche photodiode (APD) or semiconductor optical amplifier (SOA)-PIN receivers [1–3] suffer from device gain-bandwidth (GBW) product limitations [4] and severe chromatic dispersion (CD), especially for 20km fiber reach in PONs. Both non-return to zero (NRZ) and optical duo-binary (ODB) modulation format based DD scheme require heavy equalization in receiver DSP [1, 2], though ODB has better CD tolerance than NRZ. In [3], despite the use of 21-tap FFE and MLSE for CD compensation, a 7 dB sensitivity penalty at bit error rate (BER) $1e-2$ is measured after 12 km transmission under an accumulated CD value of 42 ps/nm at 1342 nm for 100 Gb/s NRZ. Alternatively, single sideband (SSB) transmission with $4 \times \lambda$ and direct detection link emerges as a potential candidate for high speed PON due to its better CD tolerance [5]. However, SSB modulation requires more complex DSP such as Hilbert transform as well as a more complicated IQ-Modulator [5] at TX side. In addition, the residual CD induced phase distortion still needs to be compensated at RX side. Coherent modulation and detection schemes offer potential high rates and robust CD tolerance via high order modulation and advanced digital signal processing (DSP) [6, 7]. But its cost, power and complexity can be prohibitive for PON deployment.

To break the barriers of IMDD methods and not be limited by the complexity and power-hungry DSP from SSB modulation and conventional coherent schemes, intensity modulation with

heterodyne detection (IMHD) receivers named simplified/quasi-coherent (S/QC) RXs have been proposed in recent years [8–11], offering a compelling scheme that combines low complexity, good sensitivity and strong CD tolerance. The IMHD scheme implements OOK modulation in TX and analog signal processing for CD suppression and demodulation in RX. In [11], instead of implementing complicated SSB modulation in TX, RX generates quadrature SSB amplitude-modulated (AM) intermediate frequency (IF) signals via heterodyne detection and the natural frequency response of the transimpedance amplifier (TIA). This approach filters out CD-distorted upper sidebands and hence preserves phase information, significantly reducing the need for CD compensation. After squaring AM I/Q signals and then adding the squared outputs, the demodulated baseband signal is enhanced while the corresponding content around $2 \times$ center frequency (f_0) of IF signal is suppressed through destructive cancellation, solving the data rate limitation due to the residual $2 \times$ frequency component in quasi-coherent receiver [8–10]. Besides, the IMHD scheme eliminates the need for complex coherent DSP, such as phase locking or carrier phase recovery.

This paper demonstrates a newly optimized PIC front end integrated with high speed TIA and analog demodulator based on IMHD scheme. Instead of using 2×4 MMI [11] which has uncontrollable phase deviation, thermal phase shifter embedded with 2×2 MMI improves phase accuracy of optical 90 degree hybrid. Furthermore, the co-designed layout of balanced PD (BPD) and waveguide fan-out matches the PIC and EIC interface, largely reducing the parasitic wirebond inductor for hybrid integration. Thanks to the improved PIC and co-designed assembly interface of PIC-EIC, higher data rate detection is obtained in measurement. Adhering to the LDPC FEC of $1e-2$ [12], 150 Gb/s NRZ is detected with -22 (B2B)/-16 dBm(20km

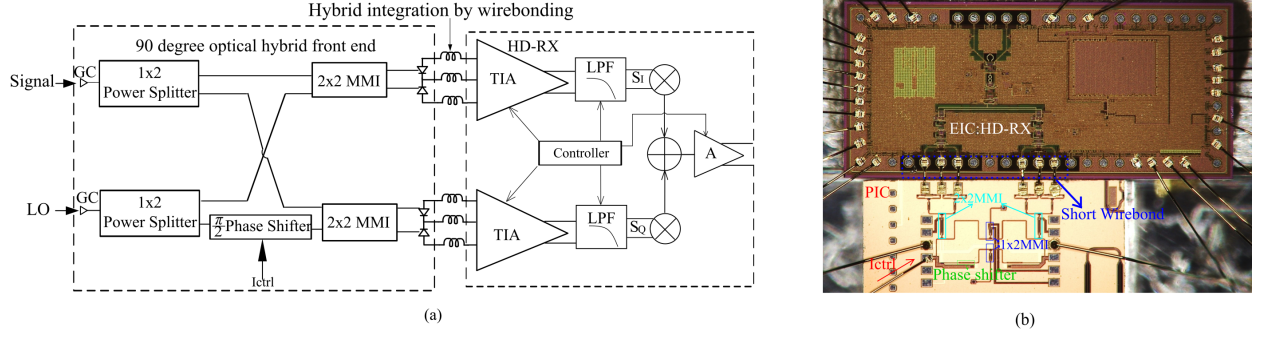


Fig. 1: (a)Block diagram of proposed HD-RX (b)PIC-EIC Assembly details

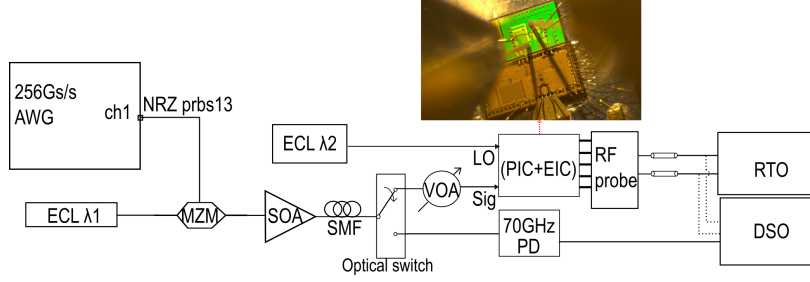


Fig. 2: Measurement setup

SMF) OMA at 1320/1295nm, using 0.5 dBm LO and 7-/11-tap FFE. No complex DSP is used.

Proposed HD-RX(PIC-EIC) and principle of IMHD

Fig. 1 (a) shows the schematic of the proposed HD RX. The received optical signal and local oscillator (LO) are coupled via a grating coupler (GC) and fed into the 90-degree hybrid optical front end. The PIC is realized by two pairs of power splitters (1x2 MMIs), 2x2 MMIs, thermally controlled optical phase shifters, and 2 pairs of BPDs. A DC current controls an optical phase shifter to achieve a 90-degree phase shift in the optical domain. Compared to directly using a 2x4 MMI to realize a 90° optical hybrid, thermo-optic phase shifters enable higher phase accuracy through post-fabrication tuning. After beating with LO, which has a wavelength offset, the received intensity-modulated optical signal (e.g., NRZ) is boosted significantly and downconverted into a pair of in-phase and quadrature (I and Q) signals around intermediate frequency (f_0). The on-chip ASP mainly consists of three functional blocks: low-pass filtering, squaring, and summation. The low-pass filtering function is realized by exploiting the frequency response of the transimpedance amplifier (TIA) as shown in Eq. (1)-(2). P_{LO} and P_S are power of local oscillator and signal respectively, R is responsivity of PD and Z_T is transimpedance of TIA. By selecting the intermediate frequency f_0 at the 3-dB bandwidth of the TIA, the upper sideband of the intermediate-frequency signal can be effectively suppressed, thereby eliminating the dispersion-induced phase distortion associated with the upper sideband. The

outputs of the in-phase (I) and quadrature (Q) TIAs are then squared using analog multipliers, followed by summation through an analog adder. In this way, the NRZ signal is demodulated back to base-band as shown in Eq.(3). Since only the phase information of a single sideband is preserved, the complexity of subsequent digital signal processing (DSP) for compensating CD induced phase distortion is significantly reduced.

Fig.1 (b) shows the hybrid integrated RX including a photonic integrated chip (PIC) fabricated in imec's iSiPP50G silicon photonic platform and an electrical integrated TIA-Demodulator fabricated in IHPSG13G2 process. The 2x2 MMI has a maximum power imbalance of 0.32 dB. The germanium photodiode has 1 A/W responsivity at a 1310 nm wavelength and a 50 GHz bandwidth with a reverse bias voltage of 3 V. The electrical TIA-Demodulator is from [11], which has a measured large signal O-E BW of 60 GHz (at medium gain settings).

$$S_I = RZ_T(f)\sqrt{P_{LO}P_S}\cos(2\pi f_0t + \phi) \quad (1)$$

$$S_Q = RZ_T(f)\sqrt{P_{LO}P_S}\sin(2\pi f_0t + \phi) \quad (2)$$

$$S_I^2 + S_Q^2 = R^2 Z_T^2 P_{LO} P_S \quad (3)$$

Measurement setup

Fig. 2 shows the experiment setup. An external cavity laser (ECL) (santec TSL-570) with 200KHz linewidth operating in the wavelength range from 1260 nm to 1360 nm is used as a laser source for the transmitter. Another ECL (Yenista TUNICS T100-HP) with 400KHz linewidth operating at same wavelength range is used as LO. The value of RIN for LO is -145 dBc/Hz in typical case.

Tab.1: Comparison with state of the art

	JLT'23[1]	JLT'25[10]	OFC'25[3]	ECOC'25[5]	ISSCC'26[11]	This work
Symbol Rate/Gbaud	50	50	100	50(x4λ)	112	150
Signal Type (Link type)	NRZ / PAM4 (IMDD)	NRZ (IMHD)	NRZ (IMDD)	NRZ (SSB-DD)	NRZ (IMHD)	NRZ (IMHD)
Sensitivity/dBm (BER 1e-2)	-23.7 / -15.4	-19.8	-22 / -18	-19.2	-25.8 / -23.7 & -22.9	-22 / -16
Fiber Reach	B2B	5km SMF	B2B / 12km SMF	20km SMF	B2B / 20km SMF	B2B / 20km SMF
Wavelength/nm	1308.7	1550	1342	1550	1310 / 1290 & 1329	1320 / 1295
DSP Complexity	13-tap FFE +1-tap DFE	No	21-tap FFE +MLSE	SSB Mod.(TX) 61-tap FFE(RX)	3-tap FFE	7-/11-tap FFE
Optical Component Responsivity/ A/W	APD 9.3	LO(10.1dBm)+PD 0.88	SOA(10dBm)+PD N.A	APD N.A	LO(4.5 / 0.41 dBm)+PD 0.9	LO(0.5 dBm)+PD 0.9

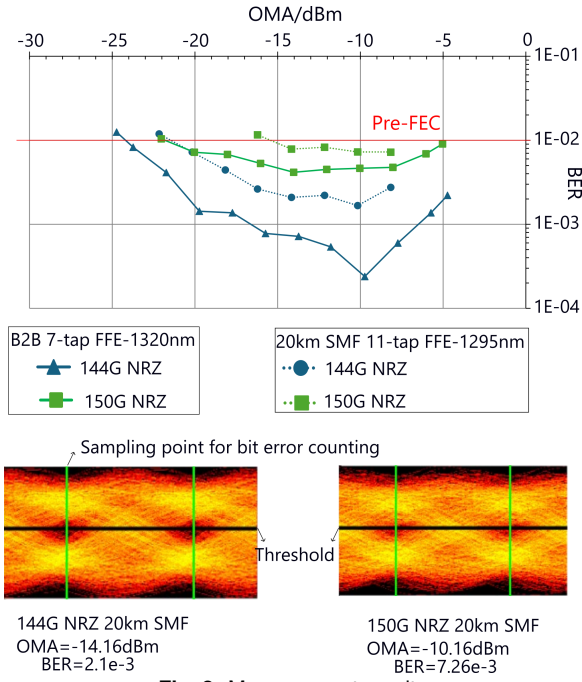


Fig. 3: Measurement results

The launching power of LO in the waveguide is set at 0.5 dBm (from 1295nm to 1320nm). An arbitrary waveform generator (AWG Keysight M8199B) with a sample rate of 256 GSa/s is used to generate pseudo random binary sequence (PRBS) data with a length of $2^{13} - 1$ at different data rates (from 144 Gb/s 150 Gb/s NRZ). The AWG directly drives a commercial Mach-Zehnder modulator (MZM), biased at its quadrature point. To investigate CD tolerance of the HD RX, 20km G.652.D SMF are inserted in the signal path. A programmable variable optical attenuator (VOA) controls optical signal power. An optical switch selects the signal, which flows either into the HD RX or into a 70 GHz bandwidth reference photodiode (PD), allowing for input eye quality checks on a digital sampling oscilloscope (DSO). The PIC is probed by two cleaved fibers, as shown in Fig. 2. The output of RX is probed by GSGSG RF probe and connected to either real time oscilloscope (RTO) for bit error counting or digital sampling oscilloscope (DSO) for output eye quality check.

Measurement results and discussion

Fig. 3 shows the bit-error rate (BER) curve for 144 Gb/s and 150 Gb/s NRZ signals under B2B and 20km SMF transmission, along with the corresponding eye diagrams. Unless otherwise specified, all reported optical modulation amplitude (OMA) sensitivities correspond to a pre-forward error correction (pre-FEC) BER level of 1E-2. For 144 Gb/s NRZ, the measured OMA sensitivities are -24.3 dBm in the B2B case at 1320 nm and -22 dBm after 20-km transmission at 1295 nm. For 150 Gb/s NRZ, the sensitivity degrades due to the limited bandwidth of both the transmitter and receiver, yielding OMA sensitivities of -22 dBm (B2B at 1320 nm) and -16 dBm (20 km at 1295 nm). Only a 7-/11-tap FFE is employed to compensate for bandwidth limitation and chromatic dispersion, without the use of more complex DSP algorithms.

Comparison with state of the art are shown in Table 1. To the best of author's knowledge, we demonstrate the highest reported data rate to date for NRZ-modulation on a single wavelength for PON downstream reception, achieving 150 Gb/s over 20 km transmission. Compared to IMDD schemes [1, 3], the proposed approach requires a significantly simpler equalizer, while operating with a local oscillator (LO) power comparable to that of the semiconductor optical amplifier (SOA). Although single-sideband modulation and direct detection (SSB-DD) scheme [5] exhibits higher chromatic dispersion tolerance, [5] requires complex single-sideband modulation at the transmitter and a long-tap equalizer at the receiver. In contrast, the proposed scheme only employs an 11-tap FFE at the receiver, without requiring any additional complex DSP.

Conclusion

A 150Gb/s NRZ heterodyne detection RX is successfully demonstrated for beyond 100G PON under 20km SMF transmission at 1295nm. The low complexity of DSP and good CD tolerance are verified, making it a promising scheme for future VHSP PONs.

Acknowledgements

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